NetrIQ.ai — SEO & LLM-Agent Visibility: Analysis and P0 Implementation

NetrIQ Engineering

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Site: https://netriq.ai · Repo: `netriq-website` (static HTML on GitHub Pages, Cloudflare DNS) · Date: 2026-05-31

1. Context — why this work

`netriq.ai` had solid on-page basics (unique `<title>` + `<meta description>` per page, `robots.txt`, `sitemap.xml`, favicon) but was effectively invisible to the two channels that now drive B2B discovery:

1. Search-engine rich results — zero structured data, no canonical tags, no Open Graph / Twitter cards, no `lastmod`. Google/Bing could not build an entity for "NetrIQ", deduplicate apex vs `www`, or render a rich snippet.
2. LLM answer engines (ChatGPT/Claude/Perplexity/Gemini search) — no `llms.txt`, no machine-readable entity graph, no preview card to cite. Asked "what is NetrIQ", an agent had only raw prose to work from.

This document records the gap analysis, the P0 fixes shipped, and the live portals + thresholds we now measure against.

2. Gap analysis (before)

Surface	Before	Impact
Title / meta description	Present, unique	OK
Canonical	None	apex vs `www` dilution, duplicate-URL risk
Open Graph / Twitter	None	no social/LLM preview card
JSON-LD structured data	None	no rich results, no entity, weak LLM grounding
`og:image`	None	no preview thumbnail anywhere
`llms.txt` / `llms-full.txt`	None	no curated source for LLM agents
`robots.txt`	Basic	didn't explicitly welcome AI crawlers
`sitemap.xml`	No `lastmod`	crawlers couldn't judge freshness
Core Web Vitals	Unmeasured	unknown ranking-factor baseline

3. What shipped (P0)

On-page metadata — every indexable page (`index`, `platform`, `verticals`, `why-netriq`, `contact`) now carries: `rel=canonical`, `robots` (`max-image-preview:large`), `theme-color`, full Open Graph (`type/site_name/locale/url/title/description/image` + dimensions + alt), and Twitter `summary_large_image`. `404.html` is set `noindex`.

Structured data (JSON-LD) — grounded only in true facts (legal entity Nysha Technologies, brand NetrIQ, `leads@netriq.ai`, +91 94825 14646, India):

- `Organization` + `WebSite` (home) — establishes the entity with logo, contact point, area served.
- `SoftwareApplication` + `BreadcrumbList` (platform) — category `SecurityApplication`, full `featureList`.
- `BreadcrumbList` + `ItemList` of the 9 verticals (verticals).
- `FAQPage` + `BreadcrumbList` (why-netriq) — backed by a new visible FAQ section so markup matches on-page content (Google policy).
- `ContactPage` + `BreadcrumbList` (contact).

LLM-agent surface — `llms.txt` (curated index per the llmstxt.org convention) and `llms-full.txt` (full distilled content for retrieval), both served from the site root.

Preview imagery — branded 1200×630 `og:image` (`assets/og/netriq-card.png`) sitewide; a product-screenshot card (`netriq-platform.png`) on the Platform page; a 512px raster logo for the `Organization` schema.

Crawl directives — `robots.txt` now explicitly allows GPTBot, OAI-SearchBot, ChatGPT-User, ClaudeBot, anthropic-ai, PerplexityBot, Google-Extended, Applebot-Extended and CCBot, and points at the sitemap; `sitemap.xml` gained `<lastmod>`.

Continuous measurement — `scripts/seo_check.py` (stdlib-only) validates every page's canonical/OG/Twitter/JSON-LD, the llms files, robots and sitemap, and — against the live URL — pulls Core Web Vitals + Lighthouse scores from the PageSpeed Insights API. Wired into `.github/workflows/seo-check.yml` (PR gate + weekly live run). Local and live runs: 41/41 checks pass.

Performance follow-through — measuring Core Web Vitals on the live site surfaced a pre-existing LCP problem (≈20 s), fixed in two further PRs:

- #26 — WebP screenshots. All product screenshots converted to WebP @1280w (PNG kept as `<picture>` fallback): 4.0 MB → 0.43 MB total (−89%); the hero `live-view` image 1.9 MB → 150 KB. Above-the-fold shot loads eager + `fetchpriority=high`; the rest lazy + low priority. Google Fonts made non-render-blocking; LCP hero image preloaded.
- #27 — hero paints at first frame. The hero (incl. the LCP element) was `opacity:0` until deferred JS ran a 0.55 s reveal, delaying the LCP paint by ~2 s after FCP. `.cr-hero .reveal { opacity: 1 }` paints it immediately; the slide-up still plays; CLS stays 0.

4. Final measured results (live, open internet)

Lighthouse (same engine as PageSpeed Insights), measured against `https://netriq.ai` after each deploy on 2026-05-31:

Metric	Before (original live)	After SEO + WebP	After hero-reveal (final)
Mobile Performance	61	84	99
Desktop Performance	79	—	83
LCP (mobile)	20.2 s	3.5 s	1.8 s
LCP (desktop)	3.2 s	—	2.5 s (green)
FCP (mobile)	3.1 s	3.1 s	1.0 s
Speed Index (mobile)	9.7 s	—	1.0 s

Metric	Before (original live)	After SEO + WebP	After hero-reveal (final)
TBT / CLS	20 ms / 0	—	0 / ~0
SEO score	(no structured data)	100	100
Best Practices	100	100	100
`seo_check.py` checks	3 / 42	41 / 41	41 / 41

Headline: LCP 20.2 s → 1.8 s and mobile Performance 61 → 99, with SEO and Best-Practices at 100 throughout. Structured data, OG cards, `llms.txt` and AI-crawler directives all verified live.

Remaining lever (not code): desktop sits at 83, now gated by origin TTFB — the page itself is optimal, but GitHub Pages is served Cloudflare-DNS-only (grey-cloud), so there is no edge CDN/Brotli/HTTP-3. Tracked as a separate infra ticket: netriq-website#28 (proxy through Cloudflare; spec in `.plans/netriq-cloudflare-proxy-perf.md`). Decision-gated on whether global/non-India traffic matters.

5. Live portals & target thresholds

These are the dashboards that expose real, third-party data for the site. Instant ones validate the moment the change deploys; accruing ones build data over days.

Instant (validate on deploy)

Portal	URL	Pass criteria
Google Rich Results Test	search.google.com/test/rich-results	Organization, SoftwareApplication, FAQ, Breadcrumb detected; 0 errors
Schema.org Validator	validator.schema.org	0 errors across all pages
PageSpeed Insights	pagespeed.web.dev	Performance ≥ 90 (mobile), SEO = 100, CWV all green (LCP < 2.5s, CLS < 0.1, INP < 200ms)
OpenGraph debugger	opengraph.xyz	Branded card renders, correct title/description
LinkedIn Post Inspector	linkedin.com/post-inspector	Card renders (re-scrape clears cache)
X Card Validator	(X dev cards tool)	`summary_large_image` renders
llms.txt reachability	`curl https://netriq.ai/llms.txt`	HTTP 200, valid markdown

Accruing (stand up now, read over days/weeks)

Portal	What it exposes	Target
Google Search Console	Impressions, clicks, avg position, coverage, rich-result enhancements	Sitemap “Success”; all 5 pages indexed; rich results “Valid”; impressions trending up
Bing Webmaster Tools	Index coverage, sitemap status (also feeds ChatGPT/Copilot grounding)	Sitemap accepted; pages indexed

Portal	What it exposes	Target
AI answer-engine spot-check	Whether netriq.ai is cited by ChatGPT / Claude / Perplexity / Gemini	“What is NetrIQ?” / “VMS for existing CCTV in India” cites or names netriq.ai after crawl

Verification cadence: instant portals at deploy and on every content change; GSC/Bing weekly for the first month then monthly; AI spot-check monthly. The GitHub Action runs the automated subset weekly and on every PR.

6. Owner actions still required (non-blocking)

1. Search Console + Bing verification — provide the verification tokens (added as `<meta>` tags in `index.html`, placeholders already in place) and add the matching DNS TXT records in Cloudflare (belt-and-suspenders, covers apex + `www`). Then submit `sitemap.xml` in both consoles.
2. **`sameAs`** profiles — confirm any public social profiles to add to the `Organization` schema (omitted for now — none confirmed).
3. Optional `PAGESPEED_API_KEY` repo secret for un-throttled CI PageSpeed runs.

7. Expected outcome

A correct entity in Google’s/Bing’s graph, eligibility for FAQ / breadcrumb / sitelink rich results, branded preview cards on every share and LLM citation, an authoritative `llms.txt` for answer engines, and — delivered — mobile Performance 99 / LCP 1.8 s with Core Web Vitals green, all continuously regression-checked in CI.